



Cybersecurity Incident of Web Cross-Site Scripting Using TCPDump and Wireshark

Group 8

Names : Muhammad Nur Falah

Naysilla Khasanah

Class : 4CS2

Faculty : Ivan Firdaus, S.T

CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2026

PENETRATION TESTING PROJECT

Batch Code : 4CS2

Start Date : 9th March, 2026

End Date : 5th April, 2026

Name of Faculty : Ivan Firdaus, ST

Names of Developer :

1. Muhammad Nur Falah
2. Naysilla Khasanah

Date of Submission: 7th April, 2026

ACKNOWLEDGEMENT

Authors would like to praise to Allah, Most Merciful bless and so authors can finish this Project 2 entitled “Cybersecurity Incident of Web Cross-Site Scripting Using TCPDump and Wireshark” Author would like to thank Mr. Ivan Firdaus, ST as Lecturer who has given useful suggestion which helps author in writing this paper.

The contents of this paper provide a detailed overview of a simple web penetration testing. The paper serves as a way of understanding penetration testing as a subject for this final semester.

SYSTEM ANALYSIS

System Summary:

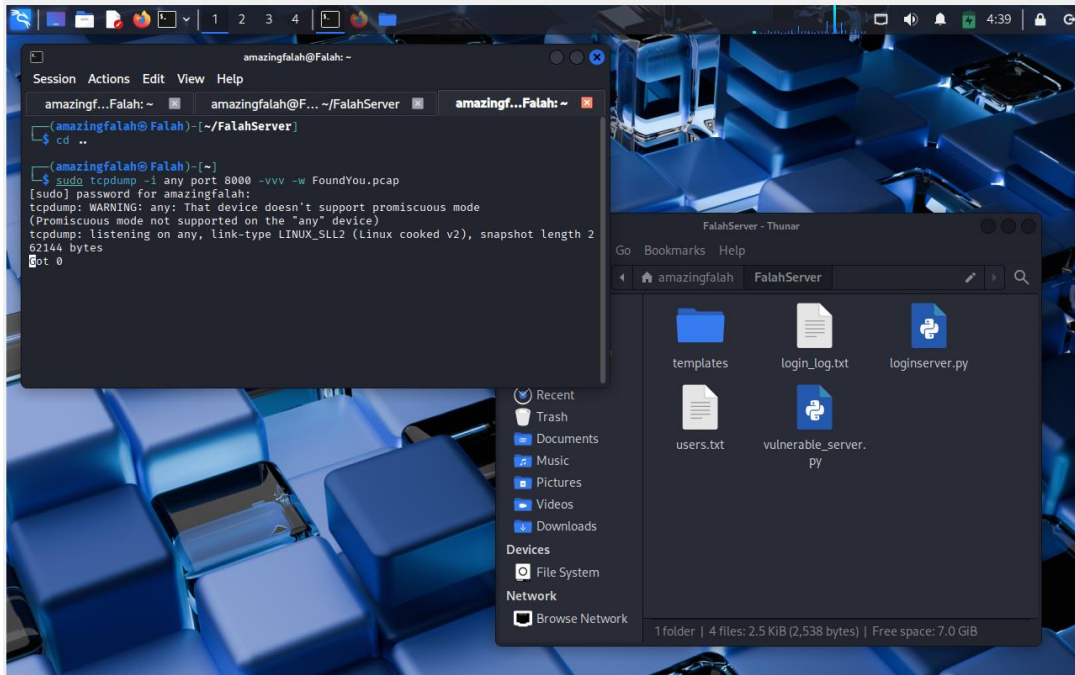
Cross-Site Scripting (XSS) are a web security vulnerability that allows an attacker to inject malicious scripts into a trusted website, which are then executed by an unsuspecting user's browser or the web itself. The browser, believing the script comes from a legitimate source, runs it with the same permissions as the site's own code, thereby bypassing security controls like the same origin policy. There are many variants of XSS, but the one our group is XSS that us to open file that we aren't supposed to have accessed to.

In this project, the penetration testing simulation focuses on grabbing important file using the web server as medium for transportation. We do this by creating a web server using python to allows any input in the URL to be execute. By doing this we created a vulnerable web server for attacker to grab the file they need. And then we can run this server in port where our TCPDump can capture the packet coming in and out.

The target machine uses Kali Linux to host a http server, while the attacker machine runs Windows 11 to open the website. Once the website fully open, the attacker can access the website URL to perform XSS. After that we can then analyse the attack using TCPDump and Wireshark tools.

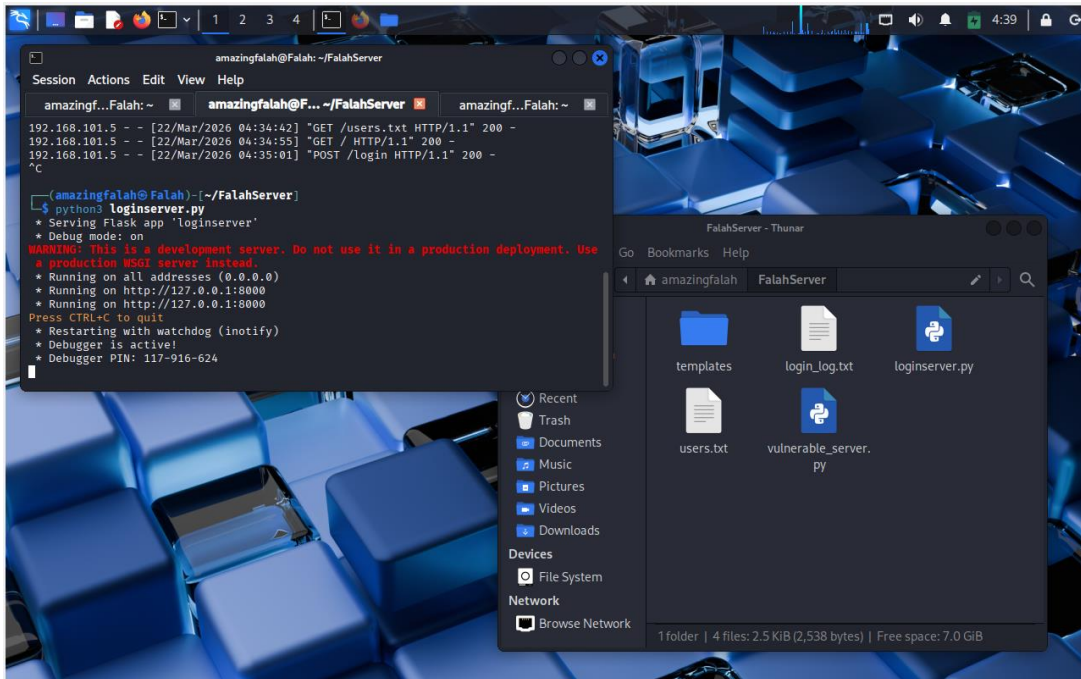
The objective of this penetration testing simulation is to demonstrate how attackers can have access to important file if our web server isn't configured properly. While also explain how to capture a packet using TCPDump and analyses them using Wireshark.

PREPARING TCPDUMP



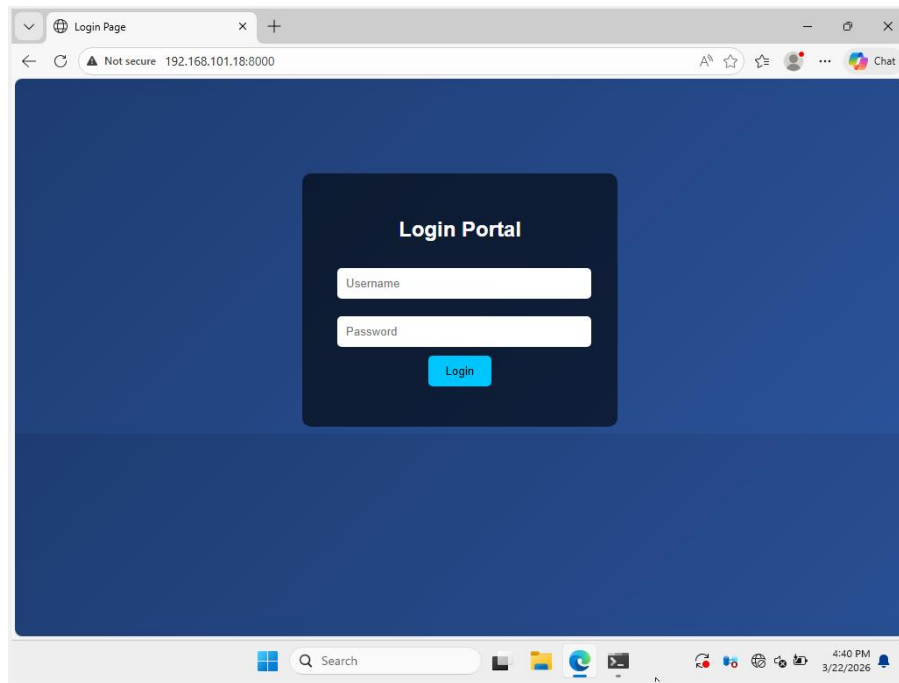
First, we need to activate our TCPDump to start capturing every packet that happen in this port. In this scenario, we use port 8000 and using any interface to capture this port. We also do “-vvv” so that the report we come with a full description. After that, we can capture and extract the data into a PCAP file, so that we can analyses the attack using Wireshark for better viewing experience.

RUN THE WEBSITE



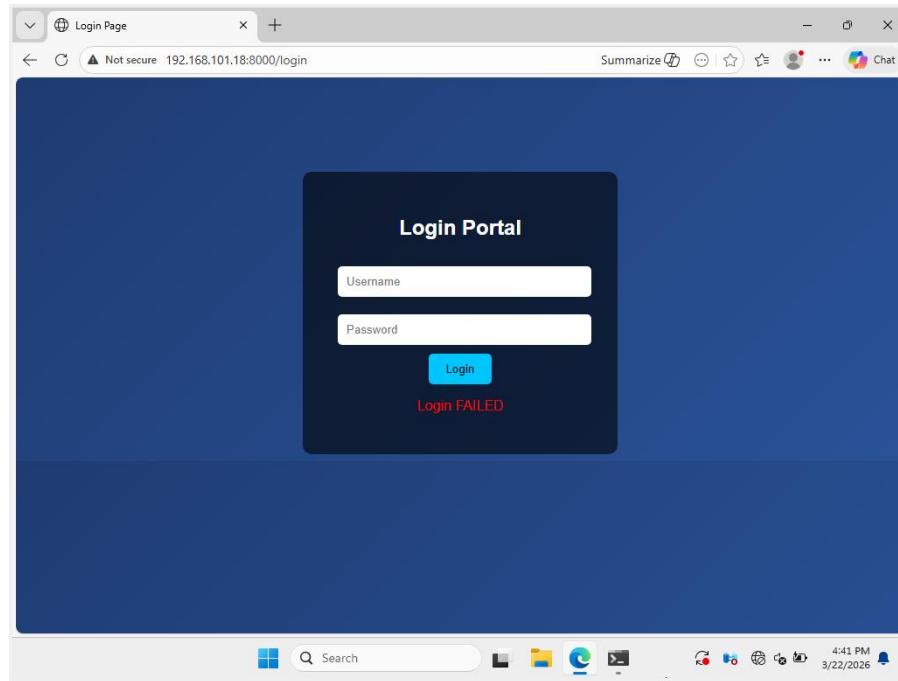
After that, we start our simulations by typing “python3 loginserver.py”. We now open our web server that are capable to log every login attempts. We can also see that in this server there are 3 important data. First is “index.html” which is located safely in “templates” folder. Second is “users.txt” that hold every valid user that can login in this web server. And lastly is “login_log.txt” that keeping track every time someone tries to login in the web server. If login is validated, it will be marked as “success” and will be marked “failed” if it’s invalid.

OPEN THE WEBSITE



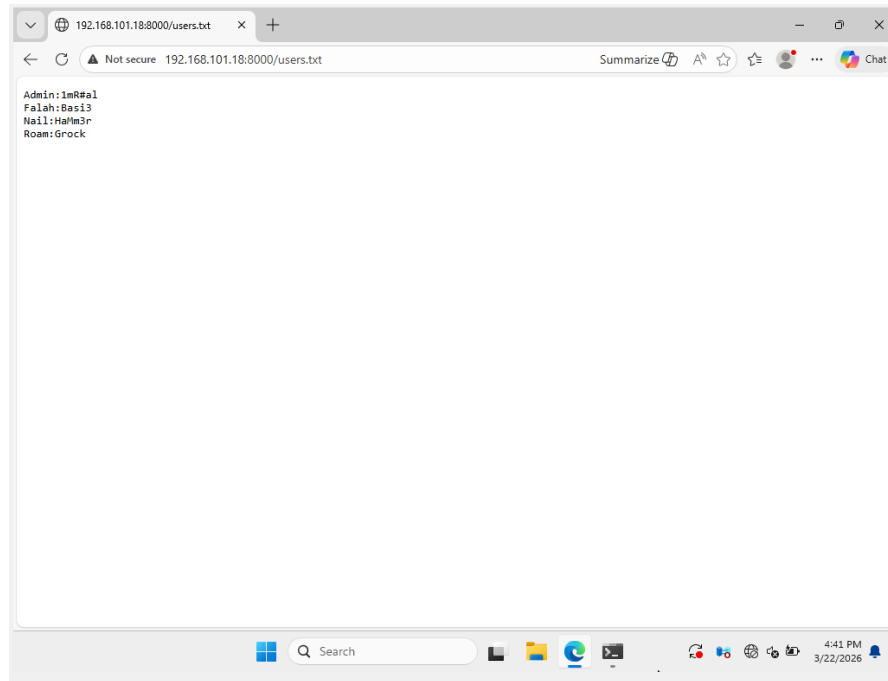
After running the web server, we can open the website by typing “http://192.168.101.18:8000”. And as we can see, it immediately opens a login page for user to input their login credentials.

BRUTE FORCE ATTEMPT



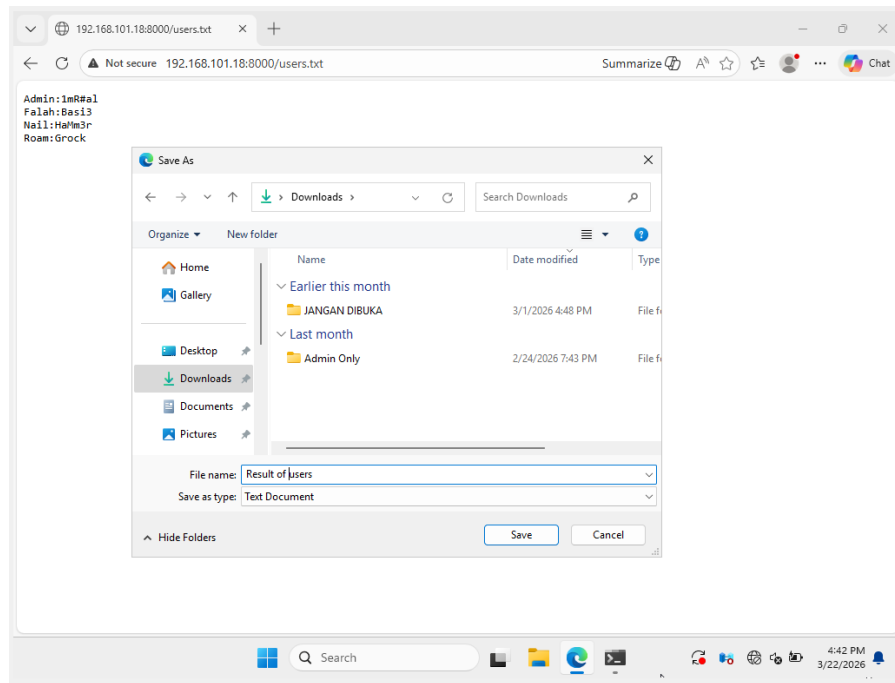
After landing on this login page, an attacker may try to brute force the login sequence. But in this scenario, the attacker unable to get a single login right.

XSS TO GRAB FILE



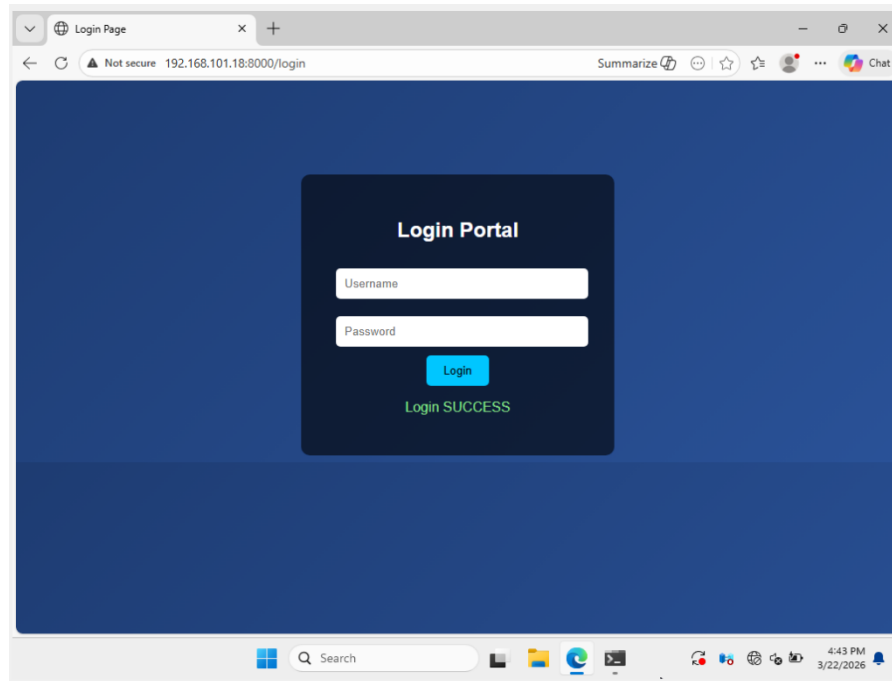
After attempting to brute force the password, the attacker then tries to attack using different approach. This time the attacker tries to open a file they aren't supposed to see using Cross-Site Scripting (XSS). By typing the file they wanted, they exploit the web server weakness that let anybody see the file that are in the same folder. After discovering this, the attacker type "http://192.168.101.18:8000/users.txt" to grab a .txt file name "users.txt". After typing this in the URL, the attacker then got what they wanted which is all the valid login credentials.

STEALING THE INFORMATION



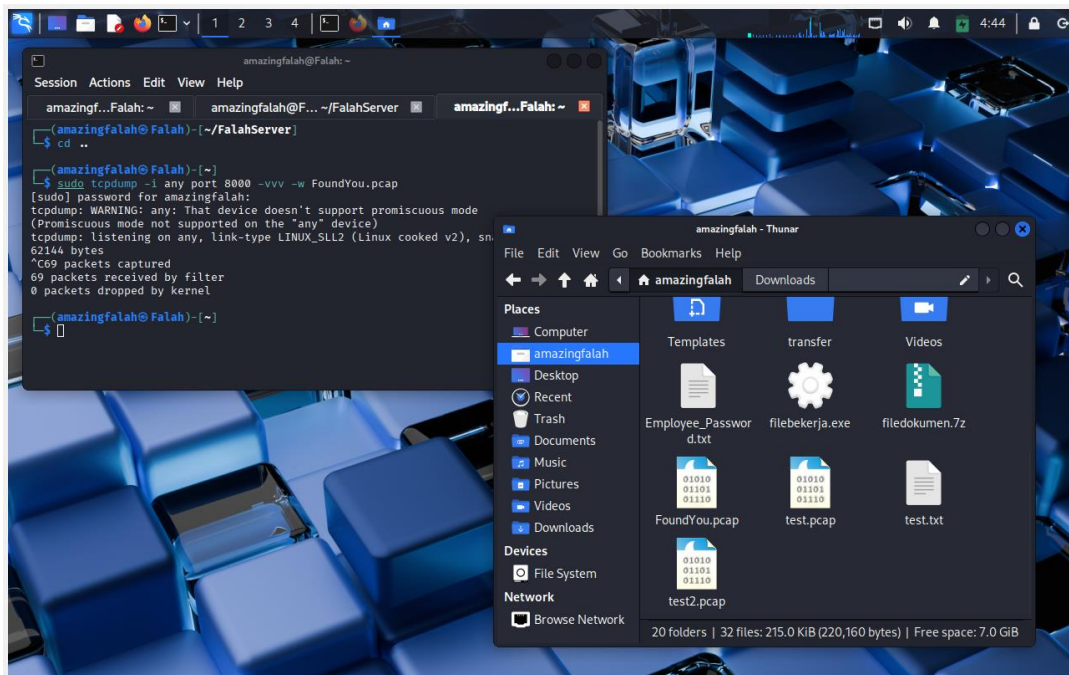
After opening the file inside the web server, the attacker can just save the stolen good using “Save As...” in web browser. This will make it easier for attacker to login in their next attempt.

LOGIN AFTER XSS



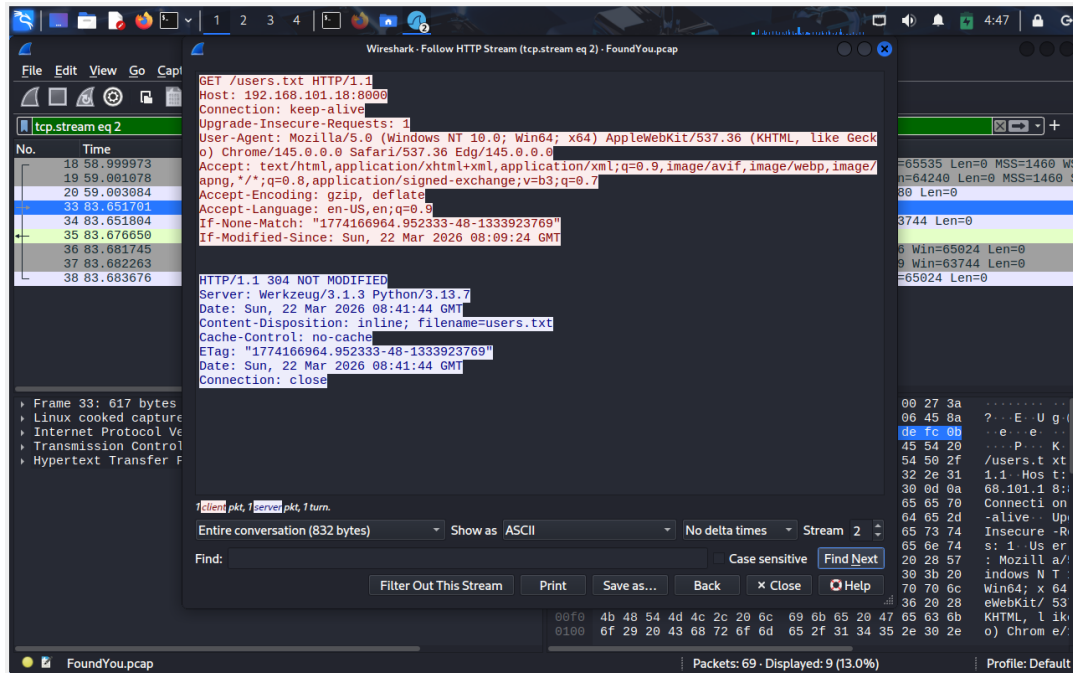
After attacker have knowledge on valid login credentials, they can now login without a worried they will be wrong. And with that, our web server has now been proven to have a vulnerable exploit that need to be patch as soon as possible.

CAPTURE OUR TCPDUMP



After we finished with attacker side of thing, we can stop our TCPDump capturing sequence by pressing “CTRL + C”. After that, we can see in our directory that a file name “FoundYou.pcap” have been created. We can then analyses this more using Wireshark for better viewing experience.

ANALYSES USING WIRESHARK



After creating “FoundYou.pcap” we can open them using Wireshark and in there we can search the packet that is the most suspect. In this scenario, the most suspect packet of all of them, are this packet that request for “users.txt”. Because normally our web server doesn’t give a button for user to just straight asking for a file, making this request the anomaly one.

CONFIGURATION

Hardware: Acer Aspire 5 14

Operating System:

1. Kali Linux
2. Windows Server 2025/ Windows 11

Software:

1. Edge
2. Terminal
3. Word

PROJECT FILE DETAILS

No	File Name	Remarks
1	Project 1 Group 1.pdf	Muhammad Nur Falah
2	Project 1 Group 1.ppt (Powerpoint, word)	Naysilla Khasanah